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Power of a point and cyclic quadrilaterals

Level 3 Senior · Geometry

Turn circle diagrams into equations using equal tangents, the power of a point and the tests for four points being concyclic.

By the end you can: use equal tangent lengths and the power of a point to turn a circle diagram into equations; prove that four points are concyclic, including with the converse of power of a point; combine power of a point with Pythagoras and similar triangles to find lengths at SMC, AMC 12 and early AIME level.

Lesson

Here is a problem to start. Two chords AB and CD of a circle cross at P. You are told that $AP = 4$, $PB = 6$ and $CP = 3$. How long is PD?

There is no radius and no angle, and yet the answer is forced: $PD = 8$. By the end of this unit that will feel obvious, and you will see that it is one face of a single fact, the *power of a point*, which also handles tangents, lines through a point outside the circle, and a surprising number of "is this quadrilateral cyclic?" questions. At SMC and AMC 12 level, circle questions that look as if they need trigonometry very often fall to two or three lines of this.

Tangent lengths

> **Equal tangents.** The two tangents from an external point P to a circle have equal length, and the line from P to the centre bisects the angle between them.

Proof: let the tangents touch the circle at S and T, and let O be the centre. A tangent is perpendicular to the radius at the point of contact, so $\angle OSP = \angle OTP = 90°$. Also $OS = OT$ and OP is shared, so triangles OSP and OTP are congruent (RHS). Hence $PS = PT$ and $\angle SPO = \angle TPO$.

The workhorse application is the incircle. If the incircle of triangle ABC touches BC, CA, AB at D, E, F, the six tangent segments come in three equal pairs: $AE = AF = x$, $BF = BD = y$ and $CD = CE = z$. Adding all six gives $2(x + y + z) = a + b + c$, so $x + y + z = s$, the semiperimeter, and therefore

$$AE = AF = s - a, \qquad BF = BD = s - b, \qquad CD = CE = s - c,$$

where $a = BC$, $b = CA$ and $c = AB$. In the same way, if a circle touches all four sides of a quadrilateral $ABCD$, then $AB + CD = BC + DA$, because each side of that equation contains each of the four tangent lengths exactly once.

> **Example 1.** A right-angled triangle has an incircle of radius 4 and a hypotenuse of length 26. Find its perimeter.
>
> *Solution.*
> The natural first move is to call the legs a and b and write $a^2 + b^2 = 676$. That is one equation in two unknowns, and the radius has not been used, so we need a way to bring the incircle into the algebra.
> Tangent lengths do exactly that. At the right angle, the incentre and the two points of contact on the legs form a square of side 4, since both radii are perpendicular to the legs. So the two tangents from the right-angled vertex have length 4. Call the tangent lengths from the other

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two vertices x and y. The hypotenuse is made of one tangent from each of those vertices, so $x + y = 26$. The perimeter consists of every tangent length twice, so it equals

$$2(4 + x + y) = 2(4 + 26) = 60.$$

We never needed a and b. For the record, $a + b = 34$ and $a^2 + b^2 = 676$ give $ab = 240$, so the triangle is 10, 24, 26, and indeed $\frac{1}{2}(10 + 24 - 26) = 4$.

The power of a point

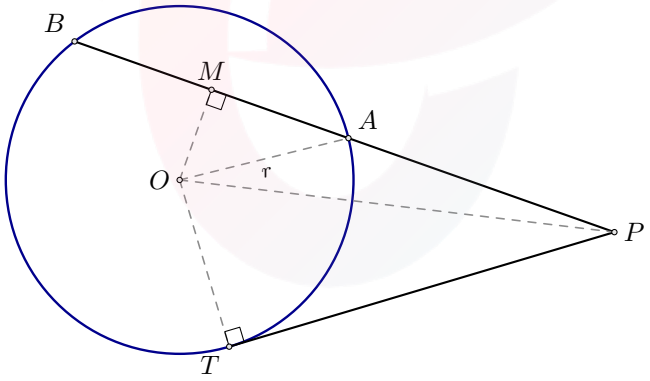
Power of a point. Let ω be a circle with centre O and radius r, and let P be any point. If a line through P meets ω at A and B, then

$$PA \cdot PB = \left|OP^2 - r^2\right|,$$

whichever line through P you choose. In particular:

- if chords AB and CD meet at a point P inside the circle, then $PA \cdot PB = PC \cdot PD$;
- if two lines through an outside point P meet the circle at A, B and at C, D, then $PA \cdot PB = PC \cdot PD$;
- if P is outside and PT is a tangent, then $PT^2 = PA \cdot PB$.

The number $OP^2 - r^2$ is called the power of P. It is negative inside the circle, zero on it and positive outside.



Proof. Drop the perpendicular from O to the line, meeting it at M. The perpendicular from the centre to a chord bisects it, so $MA = MB$. Suppose first that P is outside, as in the diagram, and label so that A is the nearer point. Then $PA = PM - MA$ and $PB = PM + MA$, so

$$PA \cdot PB = PM^2 - MA^2 = (OP^2 - OM^2) - (r^2 - OM^2) = OP^2 - r^2,$$

using Pythagoras in the right-angled triangles OMP and OMA. If P is inside, then with A again the nearer point, $PA = MA - PM$ and $PB = MA + PM$, and the same calculation gives $r^2 - OP^2$. For the tangent, Pythagoras in triangle OTP gives $PT^2 = OP^2 - r^2$ directly. The answer involves only OP and r, so every line through P gives the same product.

There is also a similar triangles proof worth knowing, because the same triangles turn up in harder problems. For chords AB and CD meeting at P, we have $\angle PAC = \angle BAC = \angle BDC = \angle PDB$ (angles in the same segment) and $\angle APC = \angle DPB$ (vertically opposite), so triangles PAC and

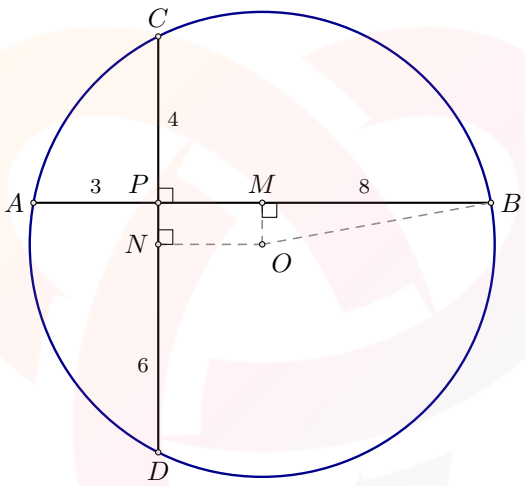
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PDB are similar and $PA/PD = PC/PB$. For a tangent, the alternate segment theorem gives $\angle PTA = \angle PBT$, so triangles PTA and PBT are similar and $PT/PA = PB/PT$.

Example 2. Point P lies outside a circle. One line through P meets the circle at A and B, with $PA = 4$ and $AB = 5$. A second line through P meets the circle at C and D, with C between P and D and $PC = CD$. Find PC, and the length of a tangent from P to the circle.

Solution.
The classic slip is to multiply PA by AB. Power of a point always measures both lengths from P: here $PB = PA + AB = 9$, so the power of P is $PA \cdot PB = 36$.
On the second line $PD = 2\,PC$, so $PC \cdot 2PC = 36$, giving $PC^2 = 18$ and $PC = 3\sqrt{2}$. A tangent PT satisfies $PT^2 = 36$, so $PT = 6$. Once you know the power of P, every line through P becomes a one-line calculation.

Power of a point gives products of lengths. To get at the radius you usually need the centre as well, and the centre is easy to locate: it lies on the perpendicular bisector of every chord.



Example 3. Chords AB and CD of a circle are perpendicular and meet at P, with $AP = 3$, $PB = 8$ and $CP = 4$. Find the radius of the circle.

Solution.
First, power of a point gives the missing length: $PD = \dfrac{AP \cdot PB}{CP} = \dfrac{24}{4} = 6$.
Now what? You could try to find an angle and use the sine rule in some triangle, but there is a cleaner observation. The perpendicular bisector of AB is perpendicular to AB, so it is parallel to CD, and the perpendicular bisector of CD is parallel to AB. So the centre O sits at a position we can read off from P without any angles.
Let M and N be the midpoints of AB and CD. Since $AB = 11$, $PM = \frac{11}{2} - 3 = \frac{5}{2}$, and since $CD = 10$, $PN = 5 - 4 = 1$. As $OM \perp AB$ and $ON \perp CD$, the quadrilateral $PMON$ is a rectangle, so $OM = PN = 1$. In the right-angled triangle OMB,

$$r^2 = OM^2 + MB^2 = 1^2 + \left(\tfrac{11}{2}\right)^2 = \tfrac{125}{4},$$

so $r = \frac{5\sqrt{5}}{2}$.
Something pretty is hiding here. Whenever perpendicular chords meet at P, we get $PA^2 + PB^2 + PC^2 + PD^2 = 4r^2$, and indeed $9 + 64 + 16 + 36 = 125$. Proving it is a good exercise: put P at the origin with the chords along the axes.

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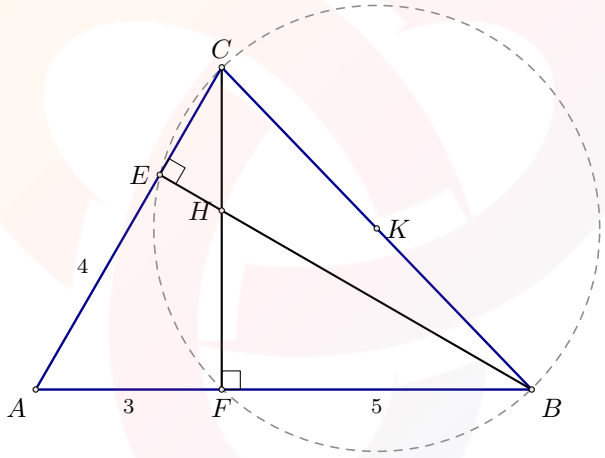
Cyclic quadrilaterals

Concyclic criteria. Four points A, B, C, D lie on a circle if any one of these holds:

1. $ABCD$ is a convex quadrilateral with a pair of opposite angles adding to $180°$;
2. C and D are on the same side of line AB and $\angle ACB = \angle ADB$;
3. (converse of power of a point) lines AB and CD meet at P, the point P lies either inside both segments AB, CD or outside both, and $PA \cdot PB = PC \cdot PD$.

You met the first two in L2-GE1. The third is the one people forget, and it is the natural partner of power of a point. Proof: let the circle through A, B, C meet line PC again at D'. Power of a point gives $PA \cdot PB = PC \cdot PD'$, so $PD' = PD$. If P is inside segment AB, it is inside the circle, so D' is on the opposite side of P from C, just as D is. If P is outside segment AB, it is outside the circle, so D' and D are both on the same side of P as C. Either way $D' = D$.

Two special cases deserve instant recall. If $\angle AXB = \angle AYB = 90°$, then X and Y lie on the circle with diameter AB. In particular, if BE and CF are altitudes of triangle ABC, then B, C, E, F are concyclic, which is the engine of the next example.



Example 4. In an acute triangle ABC the altitudes BE and CF meet at H, with E on AC and F on AB. Given $AF = 3$, $FB = 5$ and $AE = 4$, find AC and BC.

Solution.
There are right angles everywhere, so the first instinct is Pythagoras. Try it: triangle AEB gives $BE^2 = 8^2 - 4^2 = 48$, but in triangle AFC we know only AF, and nothing so far pins down C. We have plenty of right angles and too many unknowns.
The key observation is that $\angle BEC = \angle BFC = 90°$, so E and F both lie on the circle with diameter BC (dashed). Since angle A is acute, A lies outside that circle, and it has two lines through it: line AB meets the circle at F and B, and line AC meets it at E and C. Power of the point A gives

$$AF \cdot AB = AE \cdot AC, \qquad 3 \cdot 8 = 4 \cdot AC,$$

so $AC = 6$ and $EC = 2$.
Pythagoras now finishes the job: $BC^2 = BE^2 + EC^2 = 48 + 4 = 52$, so $BC = 2\sqrt{13}$. As a check, $CF^2 = 6^2 - 3^2 = 27$ and $CF^2 + FB^2 = 27 + 25 = 52$ as well.
A second route is worth seeing. In right-angled triangle AFC, $\cos A = AF/AC$, and in triangle AEB, $\cos A = AE/AB$. Equating the two gives $AF \cdot AB = AE \cdot AC$ again. Two different arguments landing on the same product is a good sign that you have found the real structure of the problem.

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Common traps.

- Measure from P. For a line through P meeting the circle at A then B, the product is $PA \cdot PB$, never $PA \cdot AB$.
- Check the order of the points before writing $PB = PA + AB$. A diagram that is not to scale can put the nearer point on the wrong side.
- Do not assume points are concyclic because the picture looks round. Prove it with one of the criteria: right angles, equal angles on the same side, or the converse of power of a point.
- In a quadrilateral with an incircle, the equal sums are of *opposite* sides: $AB + CD = BC + DA$.
- A chord through an interior point P is at least as long as the chord perpendicular to OP and at most a diameter. If a question asks for all possible lengths, check both ends of that range.

How to spot this in a contest

Reach for power of a point whenever two lines pass through a common point and both meet the same circle, especially when the question gives lengths along those lines and no angles. A tangent from an external point is the same situation with the two intersection points merged into one. If no circle is drawn, look for a hidden one: two right angles on the same segment, or two equal angles subtending the same segment, mean there is a circle in the problem whether or not the setter mentions it. Integer-length conditions, an AMC favourite, usually turn into finding a factorisation of the power. And whenever a circle touches several sides of a polygon, label the tangent lengths before doing anything else; the number of unknowns often drops from six to three or fewer.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ Chords AB and CD of a circle meet at P. Given $AP = 6$, $PB = 4$ and $CD = 11$, find the length of the longer of the two segments CP and PD.

2. ⋆⋆ Point P is at distance 13 from the centre of a circle of radius 5. A line through P meets the circle at A and B, with A between P and B, and $PA = 2\,AB$. Find AB.

3. ⋆⋆ In a convex quadrilateral $ABCD$ the diagonals meet at P, with $AP = 4$, $PC = 9$, $BP = 6$ and $PD = 6$. Given that $AB = 5$, find CD.

4. ⋆⋆⋆ A circle has centre O and radius 7, and P is a point with $OP = 3$. A chord AB passes through P, and both PA and PB are whole numbers. Find the sum of all possible lengths of AB.

5. ⋆⋆⋆ Triangle ABC has $AB = 13$, $BC = 14$ and $CA = 15$. The circle through A that touches the line BC at B meets CA again at D. Find BD.

6. ⋆⋆⋆⋆ Triangle ABC has $AB = 13$, $BC = 14$ and $CA = 15$. Its incircle touches BC at D, and the segment AD meets the incircle again at X. Find $\dfrac{XD}{AX}$.

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Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. UKMT SMC 2020 Question 15 ⋆
2. UKMT SMC 2015 Question 12 ⋆⋆
3. UKMT SMC 2016 Question 24 ⋆⋆
4. AMC 12A 2013 Problem 19 ⋆⋆⋆
5. UKMT SMC 2017 Question 19 ⋆⋆⋆
6. AMC 12B 2019 Problem 20 ⋆⋆⋆
7. AIME 1999 Problem 12 ⋆⋆⋆
8. UKMT Maclaurin 2022 Question M6 ⋆⋆⋆
9. AIME I 2024 Problem 10 ⋆⋆⋆⋆
10. UKMT Maclaurin 2021 Question M3 ⋆⋆⋆⋆
11. AIME I 2019 Problem 13 ⋆⋆⋆⋆
12. AIME I 2016 Problem 15 ⋆⋆⋆⋆

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Hints

1. Write $CP = x$ and $PD = 11 - x$. You already know the product $CP \cdot PD$.
2. Find the power of P from the distance to the centre and the radius, then write everything in terms of AB.
3. Compare $PA \cdot PC$ with $PB \cdot PD$. What does that tell you about the four vertices?
4. Power of a point gives the product $PA \cdot PB$. Then ask: what are the shortest and longest chords through P?
5. Look at the point C: one of the lines through it is a tangent. The similar triangles behind the tangent case give more than the power alone.
6. The tangent length from A to the incircle is easy to find, and then power of a point needs only one more length.

Solutions

Problem 1

Answer: 8

By the intersecting chords theorem, $CP \cdot PD = AP \cdot PB = 24$. With $CP = x$ and $PD = 11 - x$ this reads $x(11 - x) = 24$, that is $x^2 - 11x + 24 = 0$, which factorises as $(x - 3)(x - 8) = 0$. So the two segments have lengths 3 and 8, and the longer one is 8.

Problem 2

Answer: $2\sqrt{6}$

The power of P is $OP^2 - r^2 = 169 - 25 = 144$; equivalently, a tangent from P has length 12. Write $AB = x$. Then $PA = 2x$ and $PB = PA + AB = 3x$, so $PA \cdot PB = 6x^2 = 144$. This gives $x^2 = 24$, so $AB = 2\sqrt{6}$.

It is worth checking that such a line exists. The nearer intersection point is at distance between $OP - r = 8$ (the line through the centre) and 12 (the tangent), and $PA = 4\sqrt{6} \approx 9.8$ lies in that range.

Problem 3

Answer: $\frac{15}{2}$

We have $PA \cdot PC = 36 = PB \cdot PD$, and P lies inside both diagonals, so by the converse of the intersecting chords theorem $ABCD$ is cyclic. Then $\angle BAC = \angle BDC$ (angles in the same segment), that is $\angle PAB = \angle PDC$, and $\angle APB = \angle DPC$ (vertically opposite). So triangles APB and DPC are similar, with A, P, B corresponding to D, P, C. The ratio of similarity is $DP/AP = 6/4 = \frac{3}{2}$, so $CD = \frac{3}{2} \cdot AB = \frac{15}{2}$.

As a check, CP/BP should also equal $\frac{3}{2}$, and indeed $9/6 = \frac{3}{2}$. In fact that check is a proof on its own: $AP/DP = BP/CP$ with the vertically opposite angles gives the similarity directly (SAS). The converse of power of a point and this SAS similarity are the same fact seen from two sides.

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Problem 4

Answer: 27

The power of P gives $PA \cdot PB = r^2 - OP^2 = 49 - 9 = 40$.

Next, which lengths can a chord through P have? A chord at distance d from O has length $2\sqrt{49 - d^2}$, and a chord through P has $0 \le d \le OP = 3$, with every such d occurring as the chord turns about P. So the possible lengths are exactly those from $2\sqrt{40} \approx 12.6$ (the chord perpendicular to OP) up to 14 (the diameter).

The factor pairs of 40 are 1×40, 2×20, 4×10 and 5×8, with sums 41, 22, 14 and 13. Only 14 and 13 lie in the allowed range. The sum 14 is the diameter through P, with $PA = 7 - 3 = 4$ and $PB = 7 + 3 = 10$; the sum 13 is a chord with $PA = 5$ and $PB = 8$. The answer is $13 + 14 = 27$. Forgetting the diameter is the trap here.

Problem 5

Answer: $\frac{182}{15}$

From C, the line CB is tangent at B and the line CA meets the circle at D and A, so power of a point gives $CB^2 = CD \cdot CA$, that is $CD = \frac{196}{15}$. This is less than 15, so D lies between A and C.

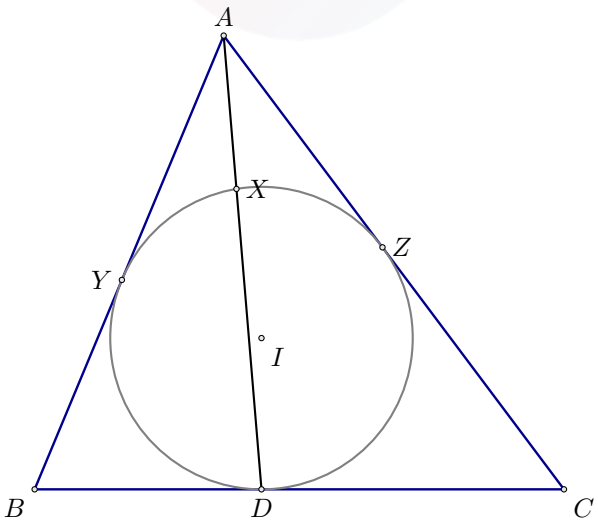
For BD, use the similar triangles behind the tangent case. The line BD separates A from C, so by the alternate segment theorem the angle between the tangent BC and the chord BD equals the angle that BD subtends at A: $\angle CBD = \angle BAD = \angle BAC$. Triangles CBD and CAB also share the angle at C, so they are similar, with C, B, D corresponding to C, A, B. Hence $BD/AB = CB/CA$, and

$$BD = 13 \cdot \frac{14}{15} = \frac{182}{15}.$$

The same similarity gives $CD/CB = CB/CA$, which is the power of a point statement again.

Problem 6

Answer: $\frac{96}{49}$



Tangent lengths first. The semiperimeter is $s = 21$, so the tangents from A to the incircle (to the points Y and Z in the diagram) have length $s - a = 21 - 14 = 7$, and $BD = s - b = 21 - 15 = 6$.

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Power of the point A with respect to the incircle gives

$$AX \cdot AD = 7^2 = 49.$$

Now we need AD, and coordinates make it painless. Put $B = (0,0)$ and $C = (14,0)$, and let $A = (x,y)$. From $x^2 + y^2 = 169$ and $(x-14)^2 + y^2 = 225$, subtracting gives $28x - 196 = -56$, so $x = 5$ and then $y = 12$. Since $D = (6,0)$, we get $AD^2 = 1^2 + 12^2 = 145$.

Therefore $AX = \dfrac{49}{\sqrt{145}}$ and $XD = AD - AX = \dfrac{145 - 49}{\sqrt{145}} = \dfrac{96}{\sqrt{145}}$, so

$$\frac{XD}{AX} = \frac{96}{49}.$$

Notice that the ratio is $\dfrac{AD^2 - t^2}{t^2}$, where $t = 7$ is the tangent length, so the awkward $\sqrt{145}$ never needed to appear.

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